

Chapter 98 — Surgical Management of Upper Urinary Tract Calculi

Robert M. Sweet, Dima Raskolnikov and Mathew D. Sorensen

CHOOSING YOUR APPROACH

American Urological Association Guidelines

- The **2016 joint American Urological Association (AUA) and Endourological Society Guideline** is the reference framework for upper tract stones, built around **Index Patients**.
- Decision making rests on **shared decision making plus a synthesis of the best available evidence**, with the caveat that endourology innovates faster than guidelines revise.
- Management is driven by three domains — **stone-related, renal anatomic and clinical (patient) factors**.

BOX 98.1 FACTORS AFFECTING MANAGEMENT OF RENAL STONES

Domain	Factors
Stone-related	Size; number; location; composition
Renal (anatomic)	Obstruction or stasis; hydronephrosis; ureteropelvic junction obstruction; calyceal diverticulum; horseshoe kidney; renal ectopia or fusion; lower pole
Clinical (patient)	Infection; obesity; body habitus deformity; coagulopathy; juvenile; elderly; hypertension; renal failure or transplant; solitary kidney; urinary diversion; pregnancy; patient symptoms

- **What the box is for:** it separates what the stone is from where it sits and who is carrying it. **Only the first domain is modifiable by choosing a different modality**; the other two constrain which modality is available at all.
- **The trap it sets:** obesity, coagulopathy and pregnancy are listed as patient factors, not contraindications — **each excludes a different modality, not surgery itself**.

Asymptomatic Renal Stones

- **Truly asymptomatic means no symptoms and no sequelae**, including no recurrent urinary tract infection (UTI). Prevalence in screened populations is approximately one in ten.
- The natural history is **not benign** — **roughly half sustain a stone-related event within five years**.
- Predictors of an event across cohorts: **number of stones, previous stone history, stone size, renal pelvis location, male sex and younger age**.
- **Lower pole stones are the commonest site observed**; over medium-term follow-up about a third enlarge, a fifth pass spontaneously and a tenth come to intervention.
- **Stone size is not a reliable predictor of symptom development**, but it does predict intervention — risk rises above 5 mm and again above 10 mm. **Size forecasts treatment, not symptoms**.
- **Residual fragments >4 mm after ureteroscopy (URS) should be treated** — higher rates of growth, complications and reintervention.
- **Sorensen et al. (2022), landmark randomised trial:** removing small asymptomatic contralateral stones at the time of surgery for the symptomatic side cut relapse to **16% versus 63%** over a mean 4.2 years.

- Large asymptomatic stones may cause **focal calyceal obstruction without symptoms** — counsel regarding gradual focal renal deterioration and infection risk.
- Non-obstructing asymptomatic stones with recurrent UTI are a special case: **the association is clear and removal is recommended irrespective of composition. Send the stone for culture** — it confirms the nidus and directs tailored antibiotic therapy.
- Where treatment is declined, the alternative is **observation with a shift to medical prevention plus serial imaging**.

KEY POINTS: ASYMPTOMATIC RENAL STONES

- Approximately half of patients will have a stone-related event over five years.
- **Number of stones, previous stone history, size and location all matter** in counselling.
- Counsel about treating contralateral stones, **even if asymptomatic**.
- **Residual stones >4 mm should be treated** after ureteroscopic treatment.
- With recurrent UTIs, **treating the stone may eradicate the infections**.

Preoperative Evaluation

- **Intravenous pyelography is largely obsolete**, displaced by renal ultrasound, plain radiography of the kidneys, ureters and bladder (KUB), and increasingly computed tomography (CT), which is sensitive and specific even at low dose.
- AUA position: clinicians **should obtain non-contrast CT before percutaneous nephrolithotomy (PCNL)**, and **may obtain it when choosing between shock wave lithotripsy (SWL) and URS**.
- Two SWL-specific CT variables: stone-free rate (SFR) is highest when **stone density is <900 Hounsfield units (HU)** and **skin-to-stone distance is <10 cm** — neither obtainable without cross-sectional imaging.
- **Functional imaging** where loss of ipsilateral function is suspected; **contrast-enhanced CT** for complex anatomy.
- **All patients require preoperative urinary testing for infection**; add full blood count and serum electrolytes for relevant comorbidity or high bleeding risk. No randomised data on timing — **urinalysis and culture within the month before surgery** is the convention.

Renal Stones: Ureteroscopy versus Shock Wave Lithotripsy versus Percutaneous Nephrolithotomy

- Governing principle: **more invasive procedures clear more stone, and that benefit must be set against operative risk**, patient goals and risk tolerance.
- Staged URS is the alternative to PCNL for large burdens — it **avoids PCNL bleeding risk at the cost of longer cumulative operating time**. PCNL bleeding risk has itself fallen with tract miniaturisation, from a historically quoted **7% to transfusion rates nearer 1%**.

MODALITY SELECTION BY STONE BURDEN AND LOCATION (AUA GUIDELINE)

Scenario	Offer	Do not offer first-line / caveat
<2 cm, non-lower pole, symptomatic	SWL or URS	Counsel: SWL fewer complications, URS higher SFR
Total burden >2 cm	PCNL first-line ; staged URS an option	SWL not first-line
Lower pole >1 cm	PCNL (highest SFR) or URS	SWL not first-line — fragments clear poorly; PCNL has higher morbidity
Lower pole <1 cm	SWL or URS	—

Scenario	Offer	Do not offer first-line / caveat
Staghorn	PCNL	—
Cannot stop antiplatelet / anticoagulant	URS	Avoid SWL and PCNL

- **Landmark — Lower Pole I (Albala et al., 2001):** for lower pole stones >1 cm, SFR was **91% with PCNL versus 21% with SWL**. This single contrast generates the whole lower pole rule.
- **The pattern across the rows:** the table is really one axis (burden) crossed by one exception (lower pole). **Everything above 2 cm and everything in the lower pole above 1 cm moves away from SWL.**

European Association of Urology Guidelines

- The European Association of Urology (EAU) framework **largely parallels the AUA**, and comparison across the AUA, EAU, National Institute for Health and Care Excellence and Urological Association of Asia guidelines found **high consensus on the surgical treatment of renal stones**.

FIG. 98.1 TREATMENT ALGORITHM FOR RENAL STONES (EAU) — MODIFIED FROM TÜRK ET AL., EUR UROL, 2017

Stone size	Order of preference
>20 mm	1. PCNL 2. URS or SWL
10–20 mm	SWL or endourology (equipoise)
<10 mm	1. SWL or URS 2. PCNL

- Applies to all but lower pole stones of 10–20 mm. Endourology here means PCNL and URS; URS includes retrograde ureterorenoscopy.
- **The contrast with the AUA table:** the EAU frames the middle band as explicit equipoise rather than a preference, which is the point of divergence to remember.

Stone Morphometry

- **Classification of stone complexity, chiefly to predict PCNL outcomes.**

COMPARISON OF STONE MORPHOMETRY SCORES

Score	Inputs	Derivation	Predicts
Guy's (2011)	Location, number, anatomy	Single centre; internally validated	SFR on KUB; not complications
STONE	Size, location <i>plus</i> tract length, obstruction, stone density	Single surgeon; all had preoperative CT	SFR, blood loss, operative time, length of stay
CROES nomogram	Size, location, number, prior treatment, staghorn, annual centre volume	Very large multinational multicentre	SFR (KUB only); argued most generalisable
S-ReSC	Distribution of stones across calices	CT before and after PCNL	SFR; not complications

- **What the table is for:** each score adds the variable the previous one lacked — anatomy, then CT-derived depth and density, then centre-level and treatment history, then calyceal distribution.
- **The pattern: three of the four predict stone clearance and none predicts complications.** Meta-analysis shows marked heterogeneity in development, validation and implementation, and **no optimised, generalisable score yet exists**.

- **The trap:** CROES is the only one with multicentre derivation, but its SFR was assessed on plain radiography alone.
- The **Triple D score** sits outside this group — it predicts **SWL** efficacy, not PCNL.

SHOCK WAVE LITHOTRIPSY FOR RENAL STONES

History

- The tissue effect of shock waves was found **by accident at Dornier**, during work on raindrop and micrometeorite damage to aircraft, when an engineer touched the shock wave apparatus.
- **First human SWL: 7 February 1980, Dornier HM1. First HM3 installations: Stuttgart 1983 and Indianapolis 1984.** The HM3 was **large, fixed and expensive but outstandingly effective with an excellent safety profile** — later generations traded **efficacy for lower cost, smaller footprint and versatility**.

Mechanisms of Stone Fracturing

- A shock wave is a **short-duration acoustic pressure wave: peak positive 30–100 MPa followed by a negative tensile tail of about –10 MPa**.
- Waves are generated at **F1 (in the therapy head)**, focused, and transmitted to the stone at **F2**. Machines differ chiefly in **peak positive pressure and the size and shape of the focal zone at F2**.
- **A focal zone larger than the stone improves breakage** — greater stress within the stone and less penalty from respiratory movement of the target.

Energy Source

COMPARISON OF SHOCK WAVE ENERGY SOURCES

Source	How it works	Strength	Weakness
Electrohydraulic ("spark gap"; original HM3)	Underwater spark generates a spherical expanding wave at F1	Very effective at breaking stones	Electrode life <40 000 shocks ; great shock-to-shock variability, focus drifting up to 1 cm as electrodes wear
Piezoelectric	Up to several thousand elements on a spherical reflector, summed at F2	Very long service life; stable focus; anaesthesia-free because skin energy density is low	Low delivered energy and very small focal volume at F2
Electromagnetic (all modern machines)	Magnetic field focused through an acoustic lens or reflector	Consistent quality up to 1 000 — 000 shocks ; broad skin distribution allows treatment without anaesthesia; high energy density at F2	

- **What the table is testing:** each source trades **focal accuracy and longevity against delivered energy**. The electromagnetic source is dominant because it is the only one that wins on both, which is why it is also the answer to "what do modern lithotripters use".
- **The trap:** piezoelectric's long life and stable focus are often misread as superiority — its delivered energy and focal volume are the smallest of the three.

Stone Comminution

- Applied shock waves generate stresses in the stone; beyond a critical value **cracks form, then grow and coalesce under repetitive stress**. Five mechanisms are proposed.

- **1. Tear and shear forces** — elastic waves generated inside the stone reflect, superimpose and focus; **cracks form where shear and tension are highest**. Once a fragment is smaller than the pulse duration, internal wave propagation, and hence this mechanism, falls away.
- **2. Spallation** — the pulse inverts on reflecting at the soft tissue or fluid interface distal to the hard stone; incident and reflected waves overlap to create **maximal tension that breaks the distal end of the stone**.
- **3. Quasistatic squeezing** — when the wave is wider than the stone it travels faster in stone than in surrounding fluid, leaving the stone in a **circumferential hoop of squeezing pressure**.
- **4. Cavitation** — the negative phase generates bubbles at the stone; their collapse **erodes the stone surface**, and this matters more as fragmentation proceeds. The paired caveat: **bubbles decay slowly, so excess cavitation shields the stone from subsequent shocks**.
- **5. Dynamic squeezing** — with a focal area wider than the stone, mechanisms 1 to 3 combine: **intense shear waves at the stone edges are reinforced by surrounding waves and focus at the distal end**, coinciding with the spallation waves. This is **the most powerful force in initial comminution**.

Factors to Improve Success and Safety

Patient Factors

- Efficacy is governed by **stone density and composition, skin-to-stone distance, stone volume, location within the collecting system, and the Triple D score**.
- Two groups to divert to a more efficient modality: **infection stones, and stones in multiple locations likely to need several sessions**.
- **Absolute contraindications in the body text** add **severe untreated hypertension** and **aortic aneurysm >4 cm** to the boxed list, and specify **current antiplatelet, antithrombotic or anticoagulant use** alongside uncorrected coagulopathy.

BOX 98.2 CONTRAINDICATIONS TO SHOCK WAVE LITHOTRIPSY

Category	Contraindication
Patient state	Pregnancy; uncorrected coagulopathy or bleeding diathesis; untreated urinary tract infection
Adjacent vascular anatomy	Arterial aneurysm near the stone (renal or abdominal aortic)
Downstream anatomy	Obstruction of the urinary tract distal to the stone
Targeting	Inability to target the stone (skeletal malformation)

- **What the box is for:** three of the four categories are about **what the shock wave will hit on the way in, or what the fragments will meet on the way out** — only the first is about the patient's physiology.

Treatment Factors

- **Coupling quality is the key technical determinant**. Apply ultrasound gel **directly from the jug onto the therapy head as a large mound** — never spread or layered.
- **Air pockets in the coupling medium are the classic avoidable error**, degrading fragmentation by up to two-fifths. Contemporary machines such as the Dornier Delta III mount **a camera in the therapy head to inspect coupling**.
- **Manufacturers: Dornier, Siemens, Storz, Wolf and EDAP-TMS**; head-to-head comparisons are few, and **most modern multifunctional lithotripters use an electromagnetic source**.

- **Shock rate:** more than a dozen randomised trials give **more than twofold better outcomes at 60–70 or 80–90 shocks per minute than at 100–120**. Fast rates accumulate a cavitation bubble cloud, causing **more tissue injury and more shielding of the stone**.
- **Ramping:** stepwise voltage escalation improves fracturing and reduces renal injury. **Slow ramping over at least four minutes, with or without a brief pause, produces detectable renal vasoconstriction early in the session**. Low-energy pretreatment markedly reduces lesion size in animals and **substantially lowers the risk of renal haematoma in humans**.
- Cavitation is essential to comminution but is **likely the greatest contributor to adverse tissue effects** — rupture of a small vessel allows blood to collect in tissue, which in turn increases further cavitation and injury.

Preparation for Shock Wave Lithotripsy

- Sequence: **history and examination → review medications and conditions for contraindications and bleeding risk → imaging to confirm the stone is visible on the targeting modality, KUB or ultrasound → urinalysis with reflex culture → treat bacteriuria or UTI before the procedure**.
- AUA and EAU agree: antibiotic prophylaxis is unnecessary with a negative urinalysis and no risk factors.

Complications

- SWL is non-invasive but has been associated with injury to **kidney, liver, skeletal muscle, colon, spleen and pancreas** — **exceptionally rare and usually not clinically significant**.
- **Cardiac arrhythmia** is common during treatment and **usually resolves with a treatment pause**; occasionally the machine must be **gated to the refractory phase of the cardiac cycle**.
- **Renal haematoma** occurs in a minority, mostly asymptomatic or mild; rarely significant enough to need **selective embolisation**. Risk factors: **increasing age, hypertension (especially poorly controlled), obesity, diabetes mellitus, coronary artery disease, thrombocytopenia, narrow-focus machines with high peak positive pressures, higher energy settings, and greater total shock number**.
- **Steinstrasse** affects a small minority overall but **rises several-fold with a stone burden above 2.0 cm²**.
- **Do not stent at the time of SWL** — no improvement in passage rates, only added stent morbidity.
- **Medical expulsive therapy with alpha-blockers** improves fragment clearance, reduces pain and shortens time to clearance. **Mechanical percussion, diuresis and inversion therapy** also improve clearance, in weekly ten-minute sessions over several weeks.
- **Refuted associations** — **worth knowing as negatives:** diabetes mellitus, renovascular hypertension, long-term renal function decline and reduced fertility.

Future Directions: Burst Wave Lithotripsy and Ultrasonic Propulsion

- Two investigational technologies on **one ultrasound platform: burst wave lithotripsy (BWL) breaks stones; ultrasonic propulsion repositions fragments**. Both are usable **awake, in clinic or the emergency department**, with **real-time ultrasound targeting** of breakage, repositioning and cavitation; the probe is handheld or support-assisted.

COMPARISON OF SWL, BURST WAVE LITHOTRIPSY AND ULTRASONIC PROPULSION

Feature	SWL	BWL	Ultrasonic propulsion
Purpose	Break stone	Break stone	Reposition stone or fragments
Pulse	Single broadband shock	5–100 cycles , narrowband burst	1500 cycles , long duration

Feature	SWL	BWL	Ultrasonic propulsion
Peak pressure	30–100 MPa positive, –10 MPa tail	4–8 MPa	2.4 MPa maximum positive
Anaesthesia	Usually required (source-dependent)	Not needed in most cases	Not needed
Cavitation	Substantial; limits repetition rate	Less than SWL; allows faster repetition	Little
Limiting effect	Tissue injury	Tissue injury (reduced)	Transducer heating and skin warming
Fragment result	Variable	Predictable, uniformly sized	Not applicable

- **What the table is for:** total energy is similar between SWL and BWL — BWL simply spreads it as **lower pressure over longer duration**, and every downstream difference (no anaesthesia, less cavitation, uniform fragments) follows from that one substitution.
- Mechanism resembles SWL, but **multiple reflections of many cycles superimpose internal waves and amplify peak negative pressure more than fivefold**, so intrastone tensile stress matches SWL despite lower pressure through tissue.
- **Coupling remains essential**, with gel; the safeguard is elegant — **if coupling is lost, the image is lost, which itself signals the user**. The **beam must be wider than the stone**, as with SWL. Excessive cavitation appears as a **dynamic white cloud on the ultrasound image**.
- Propulsion repositions by **momentum transfer when the focused wave reflects off the stone; force is directed away from the transducer**, so the probe must be manipulated to steer fragments. Uses: **accessing stones intraoperatively, and facilitating passage of ureteral stones and residual renal fragments**.

KEY POINTS: SWL, BWL AND ULTRASONIC PROPULSION

- Modern lithotripters use an electromagnetic shock wave source.
- Several mechanisms contribute to stone fracturing with SWL.
- **Stone density and composition, skin-to-stone distance, stone volume, location and the Triple D score** influence SWL efficacy.
- Absolute contraindications include **pregnancy, uncorrected coagulopathy or antiplatelet/antithrombotic/anticoagulant use, severe untreated hypertension, aortic aneurysm >4 cm, and untreated UTI**.
- Gel application, minimising air bubbles, and 60–70 or 80–90 shocks per minute increase effectiveness.
- **Renal haematoma, steinstrasse and cardiac rhythm abnormalities** are the known complications.

URETEROSCOPY FOR RENAL STONES

History

- **1806 — Bozzini's Lichtleiter ("light conductor")**, the **first clinically used endoscope**: a 35 cm tube of sheet metal, paper and leather housing a **candle-holder and mirror system**. Results were mixed, Bozzini died of typhoid in 1809, and the device was **never clinically popularised** — it survives in the William P. Didusch Museum of the AUA.

- **1826 — Ségalas' "speculum urethrocystique", the first endoscope used for urological purposes;** likewise never popular.
- **1962 — Marshall visualised a ureteral stone with a fiberoptic endoscope,** the advance that enabled miniaturisation and flexible urological endoscopes.
- **Charge-coupled device (CCD) technology,** first described in the 1970s, was refined and made cost-effective from 2005; the first commercially available endoscope with a distal digital sensor appeared in 2010. The latest iteration is the single-use digital ureteroscope.
- **The universal limitation of flexible ureteroscopes is diameter —** often exceeding the narrowest ureteral segments, especially the orifice — because the design must accommodate **both a working channel large enough for irrigation and instruments and the CCD or fiberoptic bundle.**

Preoperative Considerations

- **General anaesthesia is typical,** by endotracheal intubation or laryngeal mask airway. The trade-off as a pair: **laryngeal mask is often preferable overall, but intubation allows precise control of the respiratory cycle,** reducing renal excursion and improving lasering efficiency. URS under local anaesthesia alone is feasible.
- **Both AUA and EAU strongly recommend urinary testing for infection before elective URS.**
- **Single-dose perioperative antibiotics are required,** by institutional antibiogram. **Asymptomatic bacteriuria or very high infection risk may warrant additional pretreatment; optimal duration is not agreed.**
- **Antiplatelet and anticoagulant agents.** The AUA minimum position: **for patients who must remain on these agents, recommend URS rather than SWL or PCNL.** Over-the-counter use without clear indication should be held; recent cardiac or vascular surgery makes interruption unacceptable.
- The evidence, in order: **higher bleeding rates on anticoagulation without change in SFR or complications** in early work; **meta-analysis does show increased bleeding-related complications;** quality-collaborative data show **URS on anticoagulation is safe whereas antiplatelet use increases postoperative hospitalisation.**

Principles of Retrograde Renal Access

Wires

- Wire design balances **rigidity, hydrophilicity and cost;** choice is largely surgeon preference.
- **Both AUA and EAU recommend a safety wire for most endoscopic procedures,** though the EAU concedes the recommendation is controversial.
- The counter-evidence, as a pair: **insertion and retraction forces are higher when passing a ureteroscope alongside a safety wire,** and a multicentre study found **no outcome difference with or without one.**
- **Wires can harm:** with severely impacted stones the wire may not pass, and **forceful advancement causes submucosal passage.**

Ureteral Access Sheaths

- **Takayasu, 1974 — the first ureteral access sheath (UAS),** a "guide tube for pyeloureteroscopy" devised to negotiate **the intramural ureter,** where scopes habitually met obstruction.
- **Intrarenal pressure:** any sheath lowers intrarenal pressure versus unsheathed URS, in cadaveric and in vivo animal models, and **larger sheaths lower it more —** at presumed cost of greater injury risk.
- **The examinable disconnect: UAS demonstrably reduces intrarenal pressure, yet meta-analyses have not shown a clear reduction in postoperative infection.**

- **Dusting versus basketing** is surgeon preference; the sheath's practical advantage is **simplified repeated passes of the scope**, favouring basketing and extraction of larger stone volumes. Evidence is split — **one single series showed improved SFR with a sheath, the large CROES prospective study found no difference**, and a multicentre dusting-versus-basketing comparison found **equivalent SFR with basketing procedures substantially longer**.
- **Injury**: in a consecutive series, **UAS placement caused ureteral injury in nearly half of patients, severe (involving smooth muscle layers) in about one in eight**. **Pre-stented patients were at lower risk**, presumably from prior passive dilatation. Against this, videorecorded series with 12/14 Fr sheaths and long follow-up found **almost no de novo strictures**.
- **Why the true stricture rate is unknown**: around two-fifths of patients have **no imaging at all in the year after URS**, and many of the remainder have **KUB only, which cannot assess hydronephrosis**.

Baskets

- Variety in **size, material and shape**, with limited comparative data. Trade-off as a pair: **smaller baskets are more fragile but permit greater irrigation through the working channel**. Some designs extract faster, particularly for novices. **Selection remains surgeon preference**.

Irrigation Devices and Pressure Management

- CCD miniaturisation has shrunk working channels to **typically 3.6–4.6 Fr**, so **pressurised irrigation is usually needed** to overcome resistance and maintain vision.
- **Hand-action pump use was associated with higher postoperative fever and emergency department presentation** compared with gravity bags.
- Ranked by force generated: **gravity least, then pressurised systems and foot or hand pumps, with considerable variability**.
- **Pressure sensors embedded in a wire or in the ureteroscope** allow **real-time intrarenal pressure monitoring** — promising, but **clinical trials are needed before the data can be acted on**.

Anti-Retropulsion

- **Dretler, 2001 — the "Stone Cone"**: nitinol and stainless steel wires as an **expandable tapered cone placed cephalad to the stone** during laser lithotripsy; it effectively prevented proximal migration, with **very few reinterventions for retropulsed fragments over subsequent use**.
- Similar results with the **NTrap device** (also improved SFR) and the **BackStop thermosensitive polymer**; **modified patient positioning** also reduces retropulsion.
- **Use varies widely by country, probably on cost grounds** — in a large global study, only about one in seven cases used such a device.

Stenting

- **AUA**: consider omitting a stent after uncomplicated URS when all four conditions hold — **normal anatomy, normal renal function, no concern for ureteral injury, and no planned second-stage procedure**.
- Rationale: **multiple meta-analyses show no clear benefit of routine post-URS stenting**, against well-documented **stent-related urinary symptoms and significant quality-of-life reduction**.
- **Practice lags the evidence** — many community and academic urologists still stent routine cases.

Success Rate

- **"Success" is not a settled definition** — completely stone-free versus free of clinically significant fragments, at what interval, and on how sensitive an imaging modality.

- **Lingeman, 1986** — "**clinically significant fragments**" defined as **>5 mm**. The problem: **smaller fragments frequently become symptomatic**, requiring further procedures in up to a fifth of patients within roughly 18 months.
- Proposed remedy: **report both zero-fragment rate and adjunct residual fragment rates (>2 mm, >4 mm)**.
- **SFR is dominated by imaging modality** — CT-assessed SFR spans a very wide range, with the highest "success" in studies permitting larger residual fragments. **Non-lower pole location, smaller stone size, and postoperative ultrasound or radiography rather than CT all inflate reported SFR.**

Complications

COMPLICATIONS OF URETEROSCOPY: RATE, RISK FACTORS AND MANAGEMENT

Complication	Rate	Risk factors	Management point
Sepsis	~5% pooled	Preoperative stent; positive urine culture; ischaemic heart disease; older age; longer surgery; diabetes mellitus	Treat positive cultures preoperatively; optimal duration unagreed
Bleeding / haematoma (subcapsular, retroperitoneal)	<1%, even for stones >2 cm	Moderate–severe hydronephrosis; thin renal cortex; prolonged operative time; hypertension; preoperative UTI	Rare but potentially devastating; multidisciplinary anticoagulation planning
Ureteral stricture	~3%	Access sheath injury; impacted stone	Ultrasound is the appealing surveillance modality; most patients get no imaging
Perforation	A few per cent	Falling with miniaturisation and experience	Stop the ureteroscopy and stent (expert opinion)
Avulsion	<1%	Falling with miniaturisation and experience	Proximal ureter ; immediate percutaneous nephrostomy, delayed reconstruction

- **The pattern:** the two commonest complications are infective and mechanical, and **both are pre-empted before the scope is passed** — by treating the urine and by not forcing past an impacted stone. The rare catastrophes are the ones with a fixed protocol response.
- **The trap:** quoting a low stricture rate from series in which most patients were never imaged, or imaged by radiography alone.

KEY POINTS: URETEROSCOPY

- For patients who must stay on anticoagulant or antiplatelet therapy, **ureteroscopy is the modality of choice**.
- Ureteral access sheath use **decreases intrarenal pressure and facilitates basketing, but may increase the risk of ureteral injury**.
- A postoperative stent may be omitted when there is **normal anatomy, normal renal function, no concern for ureteral injury, and no planned second-stage surgery**.
- The size of clinically relevant residual fragments, and the optimal timing and modality of postoperative imaging, are **all matters of ongoing debate**.
- The commonest complications are **stent-related symptoms, sepsis, bleeding and ureteral injury**.

PERCUTANEOUS NEPHROLITHOTOMY FOR RENAL STONES

History

- **1976 — Fernström and Johansson:** first percutaneous stone removal under fluoroscopic guidance. Through the late 1970s and early 1980s, **Arthur Smith and Ralph Clayman with Kurt Amplatz and Paul Lange (University of Minnesota)** drove the early evolution.
- **1983 — Lawson:** first retrograde-assisted access. **1984 — Wickham:** first tubeless PCNL. **1987 — Valdivia-Uría:** first supine PCNL. **1992 —** pneumatic lithotripsy introduced (Swiss LithoClast).
- Position today: **PCNL gives higher SFRs and needs fewer procedures than SWL or URS for 1–2 cm renal stones, and is the treatment of choice for staghorn calculi and most stones >2 cm, with shorter operative times above 2 cm.**

BOX 98.3 APPLICATIONS IN WHICH PCNL CAN BE CONSIDERED FIRST-LINE THERAPY

Basis	Application
Stone burden	All renal stones >2 cm; staghorn stones; matrix stones
Lower pole obstruction	Lower pole stones >1 cm; calyceal diverticula of the lower pole; infundibular stenosis of the lower pole
Retrograde route unavailable or hazardous	Urinary diversions; inaccessible lower urinary tract; history of ureteral stricture, injury or reconstruction
Coexisting pathology	Ureteropelvic junction obstruction; infected stones in non-ambulatory patients

- **What the box is for:** grouped this way it reduces to one principle — **PCNL is first-line whenever the stone is too large, or the retrograde route too poor, for the fragments to leave by themselves.**

Preoperative Considerations

- History focused on **fitness for surgery, coagulopathy, willingness to be transfused, and reversible cardiopulmonary comorbidity.** Identifying uncorrected coagulopathy and untreated UTI is critical.
- **Short-term preoperative drainage — nephrostomy or stent —** where there have been repeated admissions for complicated UTI, sepsis or pain.
- **Stop anticoagulants in consultation with the prescriber; PCNL has been shown safe on low-dose aspirin; coordinate bridging** for high thrombotic risk.
- Examination targets **positioning: general fitness, frailty, respiratory status, body habitus, spinal curvature, and knee and hip flexibility.**
- Baseline laboratory work: **renal function, full blood count and urine culture; group and screen optional. Stress testing or pulmonary function testing** for marginal functional status.
- **Non-contrast CT of abdomen and pelvis is the gold standard preoperative study.** Read **axial, coronal and sagittal series plus three-dimensional (3D) reconstructions** for stone location and branching, renal anatomy, **infundibular lengths and angles, skin-to-stone distance, stone density in HU,** and the relationships of skin, ribs and stone to **diaphragm, spleen, liver and colon.**
- A pre-existing urogram helps select access points, but **a retrograde or antegrade pyelogram during the procedure serves the same purpose and spares intravenous contrast.**
- **3D reformatting, immersive virtual reality and 3D printing** show early promise for planning challenging anatomy; against this, retrospective data found **no added benefit of 3D reformatting over orthogonal measurements for predicting SFR.**

- **Atrophic kidney:** a diuretic renogram helps judge whether treating the unit is worthwhile — **observation for an asymptomatic stone, or nephrectomy for a symptomatic or infected stone, may be better.** The relative function cut-off is debated in the 10–20% range.

Perioperative Considerations

- **General anaesthesia is preferred;** epidural or local techniques are described. **Adding local or regional anaesthesia reduces analgesic requirement and 24-hour pain scores.**
- **Single-dose antibiotic prophylaxis is recommended, culture-specific and/or tailored to the local genitourinary antibiogram.** Evidence is emerging for **an extended course to reduce infective complications.**
- **Tranexamic acid (TXA)** — Cochrane review of ten randomised trials in adults: **no increase in thromboembolic events; probable reduction in transfusion, major and minor complications and length of stay, with improved stone-free rates,** at the cost of drug adverse effects.
- **Documented TXA adverse reactions:** seizures, headache, backache, abdominal pain, diarrhoea, fatigue, pulmonary embolism, deep vein thrombosis, anaphylaxis, impaired colour vision and other visual disturbance. Given low baseline transfusion rates in modern series, the decision is individualised.

Positioning and Approaches

COMPARISON OF PRONE AND SUPINE POSITIONING FOR PCNL

Feature	Prone (Goodwin, 1955)	Supine (Valdivia-Uría, 1987)
Calyceal access	Easiest access to posterior calices; upper pole closest to skin	Easier access to anterior calices
Working area	Large	Less working area around the flank
Respiratory / airway	Increased pulmonary capacity	Airway access retained
Fragment drainage	Against gravity	Gravity-assisted during lithotripsy
Combined retrograde route	Needs split-leg adapter	Single position, no adapter
Named modifications	Split-leg prone; hip flexion to open the retroperitoneum	Galdakao-modified supine (Scoffone); Bart modified supine (Bach)
Ideal for	Standard anatomy; upper pole access	Ileal conduit or continent catheterisable channel needing simultaneous retrograde access

- **Prone set-up as a sequence:** bolster shins and ankles so the toes float → pad the knees → keep arms no higher than 90° to the torso to prevent brachial plexus injury → axillary and pectoral rolls.
- **Bolsters horizontally across the nipple line and the anterior superior iliac spine, or two vertically** — the examinable point being that **horizontal bolsters bring the kidney down more than vertical ones.**
- **Complications specific to prone positioning:** neuropraxia and ocular injury — from insufficient facial and pressure-point padding, prolonged operative time and **rigid bolsters.**
- **Supine set-up:** ipsilateral side to the lateral edge of the table, flank elevated on a bolster under the lumbar fossa, ipsilateral arm across the chest and padded.
- **The verdict:** comparative studies conclude the choice is **mainly surgeon preference** — **but competence in several positions allows tailoring to the individual patient.**

Access Techniques

Choice of Access

- **Access is arguably the most important step of PCNL.** PCNL's share of stone cases has stayed stable at about 4.5%, while **urologist-obtained de novo access has risen to roughly a third** — and **urologist-obtained access is associated with shorter stay and lower readmission, transfusion and secondary procedure rates than radiologist-obtained access.**
- Access is tailored to **gross anatomy, collecting system configuration and stone location**, guided by **CT plus a pyelogram taken after positioning.**
- **Use ultrasound once positioned but before prepping and draping, even when fluoroscopy is the primary modality** — it delineates renal anatomy through respiration, the stones, adjacent organs, diaphragm level and overlying ribs.
- **Posterior calices with a short, wide infundibulum are ideal. Upper pole posterior access gives the most access to the rest of the collecting system in most kidneys, but chest complications rise significantly above the 12th rib.**
- **Lower pole access is safe from thoracic injury but slightly increases colonic injury risk, especially on the left.** Two further caveats as a pair: **a too-anterior lower pole is hard to reach**, and **a steep lower pole-to-renal pelvis angle makes the ureteropelvic junction (UPJ) hard to reach with rigid instruments** and antegrade treatment of upper ureteral stones difficult even with flexible instruments.
- **Renal distraction** by traction on a through-and-through wire in the lower pole, **thoracoscopic or laparoscopic direct visualisation**, and **preoperative CT-guided access** are options where safety windows are tight.

Fluoroscopy-Guided Access

- **ALARA (as low as reasonably achievable) governs.** Dose-limiting measures: **visual and tactile rather than fluoroscopic cues; fixed low mAs and kVp; a laser-guided C-arm; single pulse-per-second fluoroscopy; a urologist-controlled footpedal or designated technician; a saved pyelogram as a roadmap; last-image hold; spot films for alignment, reserving live fluoroscopy for needle advancement.**
- **Orientate first: spine across the top of the image with the collecting system aligned to it**, to avoid confusion over needle movement.

COMPARISON OF FLUOROSCOPIC ACCESS TECHNIQUES

Step	Eye of the needle (Miller, 2007)	Triangulation
Starting view	Rainbowed 15–30° towards the surgeon	Anteroposterior (AP) , needle preset at the desired angle
First alignment	Needle tip laid on skin over target, then rotated into the fluoroscopy axis — the "bulls-eye" (X-Y plane)	Rainbow 15–30° towards the operator, adjust needle
Second alignment	Rainbow away, just past neutral AP, to read the Z-axis (depth)	Rotate back to AP, adjust in the cephalocaudal plane
Advancement	Secure the plane first, then advance into the calyx	Short advancements swinging between views: mediolateral when rainbowed, cephalocaudal when AP

Step	Eye of the needle (Miller, 2007)	Triangulation
Final confirmation	Depth on the rainbowed view	In AP; rainbowing discriminates superficial from deep

- **The discriminating sign, as a pair:** on rainbowing towards the surgeon, **laterally protruding posterior calices shorten (move towards the spine), while medially splayed calices, typically upper pole, lengthen.**
- **Air after contrast identifies non-dependent calices** — anterior calices if supine. Two cautions: **air blocks ultrasound imaging, and contrast must not be injected through the access needle unless placement is certain**, since extravasation obscures the image. **A darker portion of the collecting system suggests a calyx pointing directly towards or away from the fluoroscopic head.**
- **Urine return on removing the trocar confirms puncture.** If none, **return to the original 15–30° plane and recheck that the bulls-eye was maintained. Avoid copious antegrade contrast to confirm placement** — leakage makes repeat attempts harder.
- **Around ribs:** tilt the C-arm cephalocaudally — **head caudal to angle cephalad, and the converse** — and protect the intercostal bundles by **two sequential bulls-eye manoeuvres: the first to pass the needle tip safely under and past the bundle, the second angled to the target.**

Ultrasound-Guided Access

- Principal advantages: **no ionising radiation, plus identification of bowel, spleen, liver and pleura.**
- **Curved array 2–5 MHz transducer**, depth and gain optimised per patient. **Stones are identified by posterior shadowing and twinkle artefact; posterior calices are easily seen.**
- **If images are to be saved and interpreted, use radiology-standard alignment;** otherwise orientating screen left–right to probe left–right movement may be easier. **Free-hand and needle-guide techniques** are both described.

Combined Retrograde- and Antegrade-Guided Access

- Lineage: **Lawson (1983) retrograde needle placement through a ureteroscope → Hunter (1985) series → first simultaneous retrograde and antegrade renal stone treatment (1988).**
- Technique elements: **place a UAS at the start and pass a ureteroscope through it into the collecting system**, improving drainage and flushing fragments down the ureter; **the ureteroscope lets the surgeon watch and correct the percutaneous needle in the desired calyx; a basket can grasp the percutaneous guidewire for through-and-through access; threading a 0.035 inch wire retrograde through the ureteroscope and out through the 18-gauge needle saves re-advancing the scope; tract dilatation can be watched in real time.**
- **Double bulls-eye technique, as a sequence:** first bulls-eye lines the needle up **from skin to stone** → with the needle on the stone in the target calyx, **place the retrograde scope at the ostium** → second bulls-eye lined up **from the stone or calyx to the ostium**, using the scope tip as the target → **through-and-through access obtained** → dilate to the stone.
- Further uses: **the scope itself becomes the fluoroscopic or ultrasound target when the collecting system is poorly opacified or extravasated; it reaches areas percutaneous instruments cannot, fragmenting or relocating stones and fulgurating small tumours; and retrograde basketing and laser lithotripsy at the end of the case clears calices inaccessible from the original tract.**
- Outcomes: **ureteroscopic-assisted access may lower transfusion rates, permits wire placement beyond obstruction under direct vision, and reduces fluoroscopy time and the need for multiple tracts. Meta-analysis advantages: a single position for the whole case, fewer tracts, less bleeding, fewer ancillary procedures and higher stone clearance.**

- **Disadvantages, as the counterweight: need for experienced assistance (sometimes a second surgeon), need for a split-leg table if prone, and the learning curve of prone cystoscopy and ureteroscopy.**

Conversion of Access and Tract Dilatation

- **Warn the anaesthetist before dilating** — the step is stimulating and complications begin to manifest there.
- **A through-and-through wire obviates a safety wire; otherwise place an antegrade safety wire, ideally beyond the level of obstruction.**
- **Alken metal serial dilators** — dilate even dense scar effectively and are reusable. **Amplatz serial dilators** — progressively larger **firm polyurethane dilators over an 8 Fr polytetrafluoroethylene catheter that rides a 0.035 inch wire.**
- **Balloon dilators** — the advantage is **one-pass dilatation**. Sequence: pass the balloon over the dilating wire → **position the tip fluoroscopically, under ultrasound, or under direct vision (endoscopic combined intrarenal surgery, ECIRS)** → **insufflate with contrast using a LeVein pressure syringe to 20–30 atm** → **confirm full insufflation with no waist** → **advance the preloaded access sheath over the balloon with controlled continuous back-and-forth screwing pressure, not overshooting the balloon** → **fluid return confirms position.**
- **Multiple tracts** may be needed for suboptimal angles, especially **spindly and bifid collecting systems** — but independently carry **more bleeding and transfusion**, and versus matched single-tract cases **reduce renal function, especially in chronic kidney disease. The combined retrograde and antegrade approach reduces the need for them.**

Miniaturising the Percutaneous Nephrolithotomy Tract

PCNL TRACT SIZE NOMENCLATURE

Designation	Tract size
Standard	24–34 Fr
Mini	15–22 Fr
Ultra-mini	11–14 Fr
Microperc	<10 Fr

- Tract sizes in the literature span **8–30 Fr**; the nomenclature above is contested but standard. **Jackman, 1998** — the first "mini-perc", a **13 Fr tract**, for stones under 2 cm.
- Early signal: **equivalent SFR below 2 cm; higher SFR for stones >2 cm with standard tract sizes.**
- **Mini versus standard, meta-analysis: mini is non-inferior for SFR, with smaller haemoglobin drop, fewer transfusions and shorter hospital stay, at the cost of longer operative time and poorer visualisation; adverse event profiles do not differ.**
- **Stones >2 cm, meta-analysis of randomised trials: SFR and postoperative fever equivalent; mini has longer operative time but shorter stay, and less bleeding and transfusion than 30 Fr tracts** — but every advantage of mini disappears when the comparator is a **24 Fr tract, leaving only longer operative time.** Focused 24 Fr versus mini studies are awaited.

Instrumentation for Percutaneous Nephrolithotomy

- **Rigid nephroscopes span 8–28 Fr**; because rigid instruments must pass through the scope, **the optical lens is offset.** Two standard designs: **Wickham and Amplatz**, made in versions by Storz, Wolf and

Olympus. **In both, the main channel carries the optical fibres and serves as the working channel, with irrigation flowing around the instrument.**

NEPHROSCOPE SYSTEMS: SHEATH SIZE, WORKING CHANNEL AND LENGTH

System	Outer sheath	Working channel	Working length / note
Olympus standard	22 Fr (25 Fr with continuous-flow sheath)	Main channel, offset lens	230 mm — precludes a 24 Fr access sheath ; inflow adaptor instead
Olympus mini	15 Fr	6 Fr	220 mm; single or continuous flow, offset
Storz MIP S	7.5 Fr; 11/12 dilator and sheath	2–3 Fr — laser fibres and devices <2 Fr only	15 cm and 18 cm; 6° direction of view
Storz MIP M	12 Fr; 16.5/17.5 or 21/22 Fr	6.7 Fr — instruments to 4.5 Fr	15 cm standard, 7.5 cm paediatric, 18 cm long; 12°
Storz MIP L	19.5 Fr; 23/24 or 25/26 Fr	12.4 Fr — instruments to 11.4 Fr	15 cm and 18 cm; 12°
Wolf standard	20.8 Fr and 24 Fr	Instruments to 10.5 Fr	224 mm; also a 27 Fr extra-long at 250 mm
Wolf mini	15/18 Fr sheath, 12 Fr telescope	6 Fr	225 mm; laterally offset lens
Wolf ultra-mini (UMP)	11 and 13 Fr water-jet sheaths	Inline 3 Fr telescope	260 mm

- **What the table is for:** working channel, not sheath size, decides what can be passed. **Each step down in tract size costs an instrument class**, ending with the MIP S, which takes a laser fibre and nothing else.
- **The trap:** the Storz direction-of-view figures are **degrees, not inches**.
- **Irrigant temperature.** Two studies point the same way: **room-temperature fluid causes significant falls in body temperature, late emergence, delayed recovery and postoperative shivering; body-temperature fluid shortens awakening time but was associated with a larger haemoglobin drop.** Warm the bags beforehand or use an in-line warmer.

Lithotripsy and Extraction

- Energy options: **laser, ultrasonic, pneumatic, and dual-modality lithotripters** such as the **Trilogy** and the **ShockPulse SE**.
- Extraction tools: **reusable straight graspers and three-prong grasping forceps; disposable nitinol graspers (Perc NCircle, Perc NGauge, Perc NCompass)** — valued because they are **flexible and occupy less space in the working sheath**.
- Two removal strategies as a pair: **passive irrigation to wash fragments out, and active suction, as in super-mini PCNL**.

KEY POINTS: PCNL PLANNING AND ACCESS

- Careful examination of **axial, coronal and sagittal planes and 3D reconstructions** on preoperative CT guides surgical planning.
- **Prone, split-leg prone and modifications of lithotomy** are all similarly effective, common positions.
- Urologists are **increasingly obtaining their own access** through a combination of fluoroscopy and ultrasound.

- Ultrasound **reduces radiation exposure** and demonstrates pleura, liver, spleen and bowel.
- A **combined retrograde and antegrade approach** has advantages that may lead to better operative outcomes.
- Miniaturisation **may decrease bleeding complications at the expense of efficiency and visualisation.**

The Role of Flexible Nephroscopy

- Once the bulk of stone is cleared through the access point, antegrade and/or retrograde flexible nephroscopy and ureteroscopy is recommended — it improves SFR by clearing **isolated inaccessible calices and ureteral fragments**, and **inspects for collecting system injury**. A randomised trial showed **markedly higher stone-free rates, fewer access sites and less bleeding.**
- Technical points: **use occasional fluoroscopy to orientate and target elusive densities; use the roadmap nephrostogram taken at the start of the case; when using a flexible cystoscope as a nephroscope, beware the UPJ — the ureter usually will not accommodate that diameter; use a flexible ureteroscope to clear the ureter at the end.**
- In combined cases, **one surgeon lasers and irrigates from below while the other suctions through the percutaneous tract.** Large stones basketed from difficult calices can be **brought into view for the percutaneous operator and grasped with a larger percutaneous grasper, or the retrograde ureteroscope driven out through the percutaneous sheath with the stone.**

Exit Strategies and Drainage

- **Goals: good drainage with low intrarenal pressure, and haemostasis.**
- Sequence at the end of the case: confirm the entire collecting system **and the ureter** are clear by rigid and flexible endoscopy → **back the access sheath out under direct vision, leaving the safety wire as the last hardware removed** — at minimum to the lower ureter, ideally into the bladder and out through the urethra → **if the safety wire has fallen out during the case, re-establish it endoscopically before sheath removal → observe the access site and apply direct pressure until bleeding stops.**
- In thin patients, the rib serves as a backstop against which to compress the superficial tract. Most bleeding is controlled conservatively. **Fibrin glue injection into the tract has not reduced bleeding in several randomised trials.**
- Escalation for profuse, uncontrollable tract bleeding: re-insert and insufflate the dilatation balloon, or place a Kaye tamponade catheter over the safety wire under fluoroscopy → alert the anaesthetist → arrange selective angioembolisation.
- Drainage options: percutaneous nephrostomy catheters, a double-J ureteral stent, or totally tubeless. A Foley catheter is typically left overnight either way. Small-calibre nephrostomy tubes (≤ 10 Fr) cause less pain with no increase in blood loss.

Complications of Percutaneous Nephrolithotomy

- Historically PCNL carried significantly more complications than retrograde intrarenal surgery (RIRS).
- Cochrane review of 42 trials (Söderberg et al., 2023) — the position-changing evidence: PCNL showed no effect on major complications compared with RIRS for renal stones, while reducing the need for secondary procedures. This is credited to the evolution of PCNL tools and technique.
- **Distinctive complications of renal access — uncommon or rare, but the list to know:** acute haemorrhage; delayed haemorrhage; infundibular stenosis; collecting system injury; small and large bowel injury; splenic and hepatic injury; hydrothorax and pneumothorax; postoperative systemic inflammatory response syndrome (SIRS) and sepsis; metabolic and physiological complications related

to irrigation; venous gas embolism; deep vein thrombosis; neuromusculoskeletal complications; loss of renal function; **and death**.

SPECIAL ANATOMIC CONSIDERATIONS

MODALITY ADAPTATIONS BY ANATOMIC VARIANT

Anatomy	SWL	URS	PCNL
Horseshoe kidney	Longer skin-to-stone distance; account for the <10 cm threshold	Collecting system angle makes lateral calices difficult	Access more medial and caudal → upper pole posterior calyx usually subcostal ; more colonic injury; more access failures
Calyceal diverticulum / infundibular stenosis	Ineffective unless a patent ostium is visible in an interpolar or upper pole calyx	Effective for interpolar and upper pole; identifies and dilates or incises the ostium	Best for lower pole not reachable retrograde ; mini-PCNL favourable
Pelvic kidney	Option for stones <2 cm	Option for stones <2 cm	Bony pelvis hampers access ; ultrasound guidance, escalating to CT or laparoscopic guidance
Urinary diversion	Option, dictated by stone factors	Orifice access possible but unpredictable ; loopogram or indigo carmine to locate	Supine when no transurethral access; antegrade wire access is the reliable route
Renal transplant	Find a window avoiding the bony pelvis ; prone helps	Angled catheters or wires, or Albarrán bridge with 70° lens ; shorter access sheaths	Ultrasound to avoid bowel ; balloon to 30 atm, or metal/Amplatz dilators, for the perigraft scar rind
Spinal deformity / contractures	Bony structures in the blast path ; poor fragment passage; modest SFR, repeated sessions	Good alternative; positioning and access individualised	Equal clearance, but more secondary procedures and more infectious complications

- **What the table is for:** the variant never changes which modalities exist, only which is reachable. **The anatomy alters the approach, not the choice of operation.**
- **The pattern across the rows:** in five of the six, **ultrasound or a retrograde scope is the thing that makes access safe** — the adaptations are overwhelmingly about seeing what is in the way.

Horseshoe Kidney

- **Stones occur in roughly one in five horseshoe kidneys.** URS, SWL and PCNL are all feasible with **safety and efficacy broadly comparable to normal kidneys**.
- **Concomitant ureteropelvic junction obstruction is common**, and the **course of the proximal ureter predisposes to obstruction by crossing vessels**.
- **CROES PCNL study group (Osther et al., 2011):** median operative time was longer and percutaneous access failed more often — **5% versus 1.7%** in orthotopic kidneys.
- **Flexible antegrade nephroscopy reaches the otherwise inaccessible lower medial calices.**

Calyceal Diverticulum and Infundibular Stenosis

- **Stones form within calyceal diverticula or behind a stenotic infundibulum**; diagnosis is by **intravenous urography or CT urography**.

- **Treatment trigger, as the pair: asymptomatic patients need no treatment; treat those presenting with flank pain, recurrent UTI or haematuria.**
- **Direct percutaneous puncture onto the stone is ideal but difficult** — the cavity is small and the parenchyma overlying a focally obstructing stone is often thin.
- **Retrograde-first strategy, as a sequence:** ureteroscopy identifies whether an ostium to the obstructed calyx exists → pass a wire → if angles are favourable, **laser-incise the ostium or dilate it with a small-calibre ureteral balloon** → then treat the stone retrograde, **or** obtain percutaneous access **targeting the tip of the scope or contrast passed through the dilated ostium.**
- **Fulguration of the diverticular urothelium after stone clearance is controversial** — where performed, by resectoscope with a roller ball electrode, **or by laser.**
- **Mini-PCNL versus RIRS** in a large global match-paired study: after propensity matching, **both were effective with similar stone-free and reintervention rates** — **RIRS gave the shorter hospital stay.**

Urinary Diversions

- **Modality is dictated by stone volume, location, number and type; the type of diversion dictates the approach and the positioning.**
- **In patients whose native bladder forms part of the diversion, establish whether the bladder neck has been closed** — a surgical planning point, not a detail.
- From established wire access, the surgeon may proceed **antegrade, retrograde or combined**, depending on the location and volume of stone.

Renal Transplants

- **General consensus is to remove upper tract stones in transplant kidneys**, for two reasons kept as a pair: **risk of renal failure with obstruction**, and **non-traditional presentation because the graft is denervated.**
- **Presenting signs may be only: vague graft site discomfort, fever, oliguria, haematuria and anasarca** — classic pain is absent.

Patients with Spinal Deformity or Limb Contractures

- **Assess flexibility preoperatively** — it determines the whole approach.
- Fragment passage after SWL is hampered by **lack of ambulation and aberrant renal and ureteral anatomy**; older literature shows **only modest stone-free results with frequent need for multiple sessions.**
- **The operator requirement:** be comfortable with **supine, flank and prone positioning**, and be **hypervigilant about padding pressure points wherever they fall** — **standard positioning protocols do not apply to this population.**

LAPAROSCOPIC AND ROBOTIC STONE EXTRACTION

- **Safety and feasibility are established, but it is not first-line** — the other three modalities are less invasive and highly effective.
- **Indication, per AUA: consider only when SWL, URS and PCNL have failed or are expected to fail.**
- **The special situation is concomitant ureteropelvic junction obstruction**, where a laparoscopic approach addresses **both in one operation.** The prevailing practice, however, is to **treat the stones first and reassess the junction afterwards** before committing to reconstruction.
- **Monarch (Ethicon)** — a flexible robotic platform designed to **facilitate access using an electromagnetic targeting system, driven by a game controller.** **First United States procedure by Landman at the**

University of California, Irvine, in 2023, with a clinical study underway in collaboration with Desai at the University of Southern California.

- **With continuing robot and single-port development, the role of laparoscopy and robotics in stone disease may evolve.**

Training Considerations

- Training has concentrated on **the technically demanding skill sets: upper tract ureteroscopic access and manoeuvres; three-dimensional visualisation aids using virtual and augmented reality and 3D printing; and models and curricula teaching fluoroscopic and ultrasound-guided percutaneous access, dilatation and stone treatment.**
- **Several physical and virtual reality training platforms carry validity evidence for training**, catalogued in a systematic review.

KEY POINTS: PCNL OUTCOMES, SPECIAL ANATOMY AND TRAINING

- **Flexible nephroscopy and ureteroscopy ensure a stone-free state at the end of PCNL.**
- Serious complications of PCNL are rare but may carry **significant morbidity and even mortality.**
- **Ureteroscopy and/or PCNL are good approaches** to stones in calyceal diverticula and behind infundibular stenosis.
- **Ureteral stents and small percutaneous nephrostomy tubes are both acceptable** forms of drainage after PCNL.
- **Positioning, access, modality, approach and exit strategies should all be individualised** to maximise good outcomes.
- Many **simulation-based curricula have validity evidence** for training the knowledge and skills to treat upper tract calculi.